

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination : Semester Midterm
 Duration: **1 Hour 20 Minutes**

Semester: Spring 2026
 Full Marks: **50**

CSE421 / EEE465 : Computer Networks

Answer **ALL** questions. (**Pages: 2**)

Figures in the right margin indicate marks.

Name:	ID:	Section:
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Q1 [CO1]	A company wants to add encryption to protect data sent over the Internet. At which layer of the protocol stack should encryption be implemented? And why do other layers not need to be changed?	4
Q2 [CO2]	Sadik visits an online shopping website and gets a personalized experience based on his previous activity. Later, he changes some browser settings. When he visits the website again, the personalization no longer works, even though HTTP is still used. I. What limitation of HTTP makes personalization difficult? II. Why did changing browser settings cause the personalization to stop? III. What mechanism is normally used to provide personalization?	2 + 1 + 1
Q3 [CO3] [CO2]	A university uses a proxy server with a cache hit rate of 30%. Assume: If the object is in cache, response time = 0.02 sec If not in cache, response time = 3 sec I. Calculate the average response time of a request. II. If the proxy server is removed, all requests will directly go to the original server and take 3 sec. Explain whether using a proxy server improves overall performance.	3 + 3
Q4 [CO2]	John is sending an email using Webmail. To find the IP address of mail.focus-on-your-own-exam.com, his computer sends a DNS request to the local DNS resolver. The resolver does not have the answer, so it contacts a root server, then a TLD server, and finally the authoritative server. I. What type of DNS query does the resolver use when contacting the root and TLD servers? II. What does the resolver do after receiving the final IP address? III. Which protocol is used to send the email to the mail server?	2 + 2 + 2
Q5 [CO2]	A company's internal network connects several routers in a chain: $R1 \rightarrow R2 \rightarrow R3 \rightarrow R4$. During peak office hours, traffic on R4 becomes very high, and its buffer starts to fill up. As congestion increases, upstream routers are affected, and eventually, the sending rate near the source is reduced. I. Identify the type of congestion control used in this scenario and explain briefly how this mechanism works. II. Why does the congestion effect move backward through the network?.	3 + 3
Q6	A TCP connection experiences timeout when its congestion window (cwnd) reaches 20 MSS .	4

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[CO3]	I. What are the new values of ssthresh and cwnd after the timeout? II. How does cwnd increase after this?	+ 2
Q7	Sunbeam loads two HTML pages: Page X over a persistent HTTP connection Page Y over a non-persistent HTTP connection Page X has n objects and Page Y has 22 objects, both excluding the base HTML file. The total RTT count required to load the two pages is the same. The device takes 30 ms to send a TCP request. The HTTP request and response time for an object is 60 ms.	3 + 3 + 3
[CO3]	I. Calculate the value of n .	
[CO3]	II. Calculate the total RTT time required to load Page X.	
[CO2]	III. “As the number of objects increases, persistent HTTP with pipelining becomes better than normal persistent HTTP.” Confirm or refute with justification.	
Q8	A server sends data segments using a Selective Repeat protocol. The 5th data segment has a sequence number of 12800 and an acknowledgement number of 9100. The 3rd data segment is lost and retransmitted after the 6th data segment. Each data segment (of size 200 bytes) sent by the server is requested by the client using an HTTP Request (of size 200 bytes), and the client initially has a buffer size of 3000 bytes.	4 + 3 + 2
[CO3]	I. Calculate the initial sequence numbers (ISN) of both the client and the server. II. What acknowledgement number will the receiver send before receiving the retransmitted 3rd segment? III. After receiving the retransmitted 3rd segment, determine the RWND of the receiver.	